EncoderMoE: Classifying User-Defined Gestures Using MoE Encoders and Meta-Learning

Anonymous Author(s)
Affiliation
Address
email

Abstract

Wearable gesture interfaces using surface electromyography (sEMG) and inertial measurement units (IMUs) offer compelling pathways for accessible human-computer interaction, yet generalizing to novel users remains a fundamental challenge. This problem is especially pronounced for user-defined gesture systems, where physiological heterogeneity across individuals is compounded by semantic heterogeneity: different users may perform entirely different motions for the same gesture label. This challenge is further amplified for users with motor impairments, whose signal characteristics diverge substantially from those of unimpaired users on which most prior work is trained. We formulate user-defined multimodal gesture recognition as a few-shot cross-subject classification problem and propose **EncoderMoE**, combining Model-Agnostic Meta-Learning (MAML++) with an encoder-level Mixture-of-Experts (MoE) module. MAML++ learns parameter initializations that adapt to a new user from as few as one labeled example per gesture; encoder-level MoE provides representational diversity across specialized sub-encoders to capture heterogeneous physiological and semantic structure across diverse disability profiles. We evaluated 32 participants, 26 with upper-body motor impairments, performing user-defined executions of 10 gesture prompts via a 28-sensor EMG-IMU setup. EncoderMoE achieves 86.7% accuracy on 1-shot 3-way classification (chance: 33.3%), outperforming subject-specific transfer learning by 14.5 points. Ablation studies confirm that meta-learning and representational specialization are jointly necessary, with parameter-matched controls ruling out model capacity as a confound. These results demonstrate that pairing fast adaptation with heterogeneity-aware representations is an effective strategy for personalized biosignal interfaces for users with motor impairments.

1 Introduction

Gesture-based interaction enables intuitive, naturalistic control of smartwatches [26], extended reality (XR) systems [27], assistive robots [29], and personal computing devices [19]. User-defined gestures further enhance this potential, allowing individuals to customize gestures to fit their specific preferences and needs. This is especially critical for users with motor impairments who may fatigue easily, have limited ranges of motion, or lack fine manual dexterity due to motor disabilities like stroke, spinal cord injury, Parkinson disease, or essential tremor. Wearable sensing modalities, including surface electromyography (EMG), which captures muscle activation non-invasively, and inertial measurement units (IMUs), which capture kinematic motion data, offer compelling signal sources for customized and accessible gesture interactions. Critically, pairing EMG with IMU is not merely additive: EMG-only systems are susceptible to muscle fatigue, electrode shift, and reduced signal quality in populations with motor impairments, while IMU-only systems cannot capture

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

low-movement or fully static gestures [18, 6, 27]. Hybrid EMG-IMU interfaces overcome these complementary failure modes [18].

The central obstacle to realizing this potential is high inter-subject heterogeneity. EMG and IMU signals vary substantially across individuals due to differences in anatomy and motor strategy [3, 7]: this is especially pronounced for users with motor impairments, whose movements and signal profiles often diverge fundamentally from non-disabled populations [27]. User-defined gesture systems compound this challenge further: unlike standard classifiers, which assume a shared movement distribution, user-defined systems allow each participant to perform whatever motion is comfortable, introducing *semantic* heterogeneity on top of physiological heterogeneity. For the same gesture label, one user's motion may be entirely unrelated to another's: for example, a user with limited upper-extremity mobility may crane their neck to indicate "Select", rather than pointing [27]. Acquiring sufficient per-user labeled data is often impractical and fatigue-inducing, making rapid few-shot adaptation to each new user's personal gesture vocabulary a core requirement. Standard supervised training does not optimize for this: minimizing cross-entropy over training users encourages a global classifier that performs well for the average user, rather than reaching a parameter state that enables fast, user-conditional adaptation from a minimal support set.

Prior approaches have addressed pieces of this problem but not their intersection. Transfer learning [7], metric-learning [21], and large-scale pretraining [11] improve cross-subject generalization but struggle in our target 1-shot regime and do not accommodate personalized gesture sets. Crucially, these models often rely on global feature representations of normative physiological patterns that do not generalize to the highly idiosyncratic signal signatures characteristic of diverse motor disabilities, such as muscle spasticity or reduced range of motion. Gesture customization approaches [19, 28, 26] are limited to non-disabled cohorts or do not scale well with additional users. No prior work addresses the intersection of few-shot cross-subject generalization, user-defined gesture semantics, and multimodal EMG-IMU fusion for users with motor impairments.

We formulate user-defined multimodal gesture recognition as a few-shot cross-subject classification problem and propose a model combining two complementary inductive biases. First, we employ Model-Agnostic Meta-Learning (MAML) [2, 9] to learn parameter initializations that adapt rapidly to a new user from $K{=}1$ labeled examples per gesture. Second, we augment this with a Mixture-of-Experts (MoE) module [20] at the encoder level. In our user-defined gesture, user identity is not a nuisance variation to be suppressed: a user's signal profile and ability level directly inform gesture classification, since two users may produce physically similar movements for different gesture labels (and vice versa). Furthermore, we are motivated by the empirical finding that per-sample encoders outperform a global shared encoder in this setting [28]. MoE resolves the tension between per-sample personalization and cross-user generalization by learning to route each sample over a bank of specialized encoders, capturing inter-subject variability without collapsing the user-conditional structure essential for personalized gesture semantics. MAML and MoE are synergistic: MoE provides the representational diversity needed to span the heterogeneous signal space, while MAML learns to rapidly specialize this representation from a minimal support set.

We evaluate on a dataset of 32 participants, 26 of whom have an upper-body motor impairment, performing user-defined executions of 10 gesture prompts via a 28-sensor multimodal setup (16 EMG, 12 IMU). Our approach achieves 86.7% accuracy on 1-shot 3-way classification, outperforming subject-specific models, standard transfer learning, and state-of-the-art pretrained EMG models [11].

Contributions.

1. **Method.** A MAML+MoE architecture with encoder-level representational specialization, with an inductive bias argument for why this combination addresses compounded physiological and semantic heterogeneity of our EMG-IMU gesture recognition.

2. **Empirical Analysis.** Parameter-matched ablations against subject-specific and cross-subject baselines, demonstrating that the combination of MAML and MoE is necessary in this high-heterogeneity regime.

3. **Routing Analysis.** Visualization of MoE routing behavior across ability levels, offering mechanistic insight into how the model develops specialized expert pathways for different movement strategies and disability profiles.

2 Related Work

EMG and IMU Gesture Recognition. Cross-subject generalization remains a significant hurdle for EMG-based interfaces due to physiological variance [3]. Several studies have utilized transfer learning from large-scale pooled data to mitigate this. Côté-Allard et al. [7] demonstrated that deep convolutional architectures can learn domain-invariant features from wrist-band EMG, though this approach was limited to participants without motor impairments and required significant pre-training data. Kaifosh and Reardon [11] further scaled this to $N > 4,800$ participants, showing that while generalization is possible at scale, a subset of users still requires personalized calibration to achieve functional accuracy. Xu et al. [26] demonstrated gesture customization from limited demonstrations on wrist-worn devices, but requires a fixed gesture vocabulary and has not been evaluated under motor-impairment heterogeneity. While hybrid EMG-IMU systems leverage the complementarity of neuromuscular intent and physical kinematics [18], most models assume a standard range of motion. For instance, Tong et al. [22] utilized video embeddings to enable zero-shot IMU-based recognition, yet this relies on a "typical" movement-to-signal mapping. Such assumptions often fail for users with motor impairments, whose movements are highly heterogeneous and unique [27]. Recent efforts to customize gestures for this population, such as Yamagami et al. [28], use per-sample linear encoders followed by k-nearest neighbors to avoid overfitting, but the linearity of this encoding limits expressivity for complex multimodal biosignals.

Few-Shot Learning and Meta-Learning. Metric-learning approaches, such as Matching Networks [24] and Prototypical Networks [21], learn an embedding space where novel classes are identified via nearest-neighbor or prototype-based classification. These methods can be fragile with 1-shot physiological signals: a single EMG sample is a highly stochastic realization of a user-specific distribution, making it difficult to build reliable class prototypes [21]. Optimization-based meta-learning, exemplified by MAML [9] and its variant MAML++ [2], instead learns a parameter initialization that allows for fast gradient-based adaptation. This approach has been successfully applied to EMG recalibration for prosthetics [17] and stroke rehabilitation [13], where labeled data per session is scarce. Our work extends these techniques to the more complex setting of cross-subject generalization for multimodal EMG-IMU signals under user-defined gesture semantics.

Mixture of Experts (MoE). Shazeer et al. [20] introduced sparsely-gated MoE with top-k routing, demonstrating that conditional computation can significantly increase model capacity without proportional increases in inference cost. The Switch Transformer [8] simplified this to top-1 routing with load-balancing losses to ensure expert utilization. While MoE is traditionally associated with large language models, its capacity for representational specialization generalizes well to multimodal time series with diverse input patterns [4]. In our setting, encoder-level MoE is motivated by the finding that specialized per-sample encoders outperform monolithic models for user-defined gestures in motor-impaired populations [28], enabling the system to route each user's unique movement profile to the most appropriate internal representation.

3 Method

We propose **EncoderMoE**, combining a Mixture-of-Experts encoder bank with MAML-based meta-learning to address two challenges simultaneously: high inter-subject variability in multimodal biosignals, and the need to rapidly adapt to a new user from $K = 1$ labeled examples per gesture. The MoE encoder provides a structured hypothesis space over user-specific signal morphologies; MAML ensures the routing and representations are initialized to adapt efficiently at test time.

Task Formulation. We formulate cross-subject EMG-IMU gesture recognition as a few-shot classification problem under a strict user split protocol. Let $\mathcal{U} = \{u_1, \ldots, u_N\}$ be the set of users, partitioned into disjoint meta-train, meta-val, and meta-test splits such that no user appears in more than one split. Each user u_i possesses a personalized gesture vocabulary $\mathcal{G}_i$; while gesture *labels* are shared across users, gesture *execution* is user-defined, producing high inter-subject variability in the signal space. We sample *episodes* from a task distribution $p(\mathcal{T})$, where each task $\mathcal{T}_i$ is drawn from a single training user and consists of a support set $\mathcal{S}_i = \{(\mathbf{x}_j, y_j)\}_{j=1}^{NK}$ and a query set $\mathcal{Q}_i = \{(\mathbf{x}_j, y_j)\}_{j=1}^{NQ}$. At meta-test time, an unseen user $u_t \notin \mathcal{U}_{\text{train}}$ provides K labeled examples per

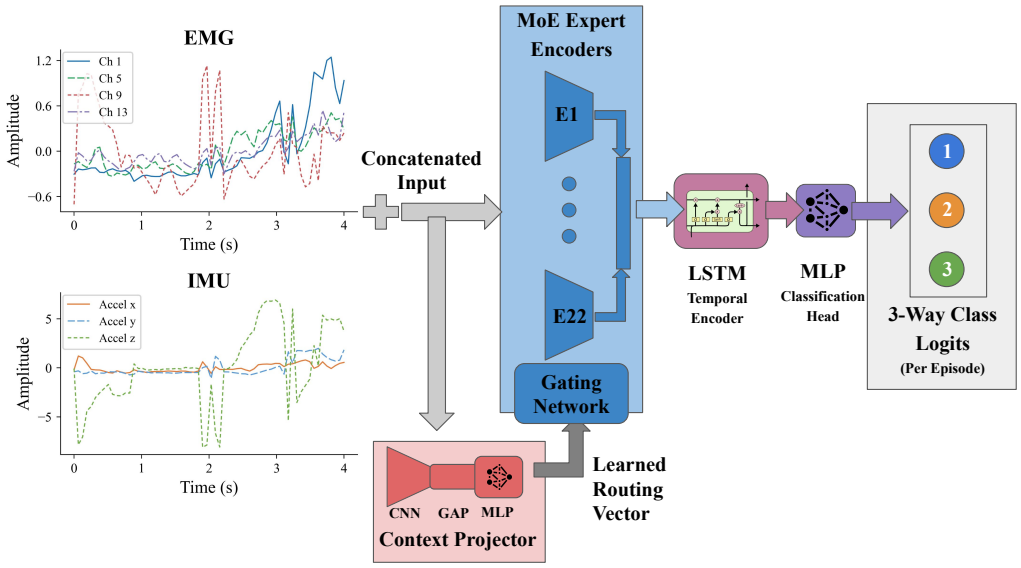


Figure 1: **EncoderMoE model architecture.** EMG and IMU data are concatenated and passed in parallel to the MoE encoder and the context projector. The context projector's learned routing vector is used by the gating network to weight expert outputs (top-k routing). Output from the MoE is passed through a shared 3-layer LSTM and an MLP classification head. Crucially, the routing signal is derived from the input itself rather than any user label, allowing the gate to discover user-specific routing without supervision (Section 4.6).

gesture from an N-way subset of $\mathcal{G}_t$; the model adapts to the support set and classifies the held-out query set.

Meta-Learning Objective. We train EncoderMoE with MAML++ [2], an optimization-based meta-learning algorithm that seeks a parameter initialization θ from which a small number of gradient steps on a new user's support set yields strong query-set performance. An inner loop adapts θ to each task via M gradient steps, producing task-specific parameters ϕ_i; the outer loop updates θ by minimizing query-set loss across tasks. This bi-level optimization that explicitly trains for fast adaptation rather than average-user performance. MAML++ extends vanilla MAML with per-parameter learned step sizes (LSLR) and a multi-step loss (MSL); full equations are provided in Appendix A.1.

3.1 Architecture

Our model combines MoE spatial CNN encoders with a shared temporal encoder and classification head to support diverse user-defined gesture classification.

Input and Modality Fusion. Each input sample is a synchronized pair of surface EMG ($\mathbf{x}_{\text{EMG}} \in \mathbb{R}^{16\times T}$) and IMU ($\mathbf{x}_{\text{IMU}} \in \mathbb{R}^{72\times T}$) recordings, resampled to T=64 time steps. The two modalities are concatenated along the channel dimension to form a single input tensor $\mathbf{x} \in \mathbb{R}^{88\times 64}$ that is processed jointly by all downstream modules.

EncoderMoE. The encoder consists of E=22 structurally identical expert CNNs $\{f_{\theta_e}\}_{e=1}^{E}$. Each expert is a 3-layer 1-D convolutional layer with a doubling filter schedule: layer ℓ has $64 \cdot 2^{\ell-1}$ filters ($\ell = 1, 2, 3$), a kernel of size 5 with same-padding, Group Normalization [25] with 8 groups, and GELU activations. Group Normalization is used throughout in place of Batch Normalization because it is stable at the small per-episode batch sizes inherent to few-shot learning. Each expert maps $\mathbf{x}$ to a feature map $\mathbf{h}_e \in \mathbb{R}^{256\times T}$.

A lightweight *context projector* reads $\mathbf{x}$ to produce a routing vector $\mathbf{r} \in \mathbb{R}^{64}$, independently of the expert CNNs. It applies a single 1-D convolutional layer (32 filters, kernel 5, GN, GELU), global

average pooling (GAP), and a two-layer MLP (hidden dim 32, output dim 64). A gating network G maps $\mathbf{r}$ to a distribution over experts:

$$\mathbf{w} = \mathrm{Softmax}(\mathbf{W}_g \, \mathbf{r} \, / \, \tau) \, , \tag{1}$$

where $\mathbf{W}_g \in \mathbb{R}^{E \times 64}$ and τ=1.53 is the temperature. We apply Top-k routing (k=9): all but the k largest weights are zeroed and the remainder renormalized to $\hat{\mathbf{w}}$. The mixture representation is:

$$\mathbf{h}_{\text{mix}} = \sum_{e=1}^{E} \hat{w}_e \cdot f_{\theta_e}(\mathbf{x}). \tag{2}$$

Temporal Encoding and Classification Head. $\mathbf{h}_{\text{mix}} \in \mathbb{R}^{256 \times T}$ is passed to a stack of three bidirectional Long Short-Term Memory (LSTM) layers (hidden size 64; output dimension 128), each followed by dropout (0.1). We prefer LSTMs over attention-based temporal models because the recurrent inductive bias imposes a stronger prior on temporal ordering, which we hypothesize reduces the degrees of freedom MAML's inner loop must adapt from a single support example. The final representation is mean-pooled across time and classified by a two-layer MLP head ($128 \to 64 \to N$).

3.2 Training

MAML++ Training. All components (EncoderMoE experts, routing network, temporal encoder, and classification head) are trained end-to-end with MAML++. Key hyperparameters selected by Bayesian optimization (Appendix A.2): E=22 experts, top-k=9 routing ($\approx 41\%$ utilization), gate temperature τ=1.53, M=10 inner steps, outer learning rate β=1.95×10^{-4}, weight decay 8.87×10^{-4}. We use first-order MAML throughout [15, 2]. Hyperparameter optimization found MSL and LSLR at training time (but not eval time) beneficial; other MAML++ components did not improve performance. The outer optimizer is Adam [12]; we train for 23 epochs of 500 episodes (meta-batch 24) with early stopping.

Auxiliary Load-Balancing Loss. To prevent *expert collapse*, where the gate converges to routing all inputs through a single expert, we add a Switch Transformer-style load-balancing loss [8] to the outer-loop meta-objective. For a batch of B samples and E experts with top-k routing:

$$\mathcal{L}_{\text{aux}} = \lambda_{\text{sw}} \cdot E \sum_{e=1}^{E} f_e \, P_e, \tag{3}$$

where $f_e = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}[e \in \text{top-}k(\mathbf{w}^{(b)})]$ is the mean dispatch fraction (detached from the gradient graph) and $P_e = \frac{1}{B} \sum_{b=1}^{B} w_{\text{soft},e}^{(b)}$ is the mean soft routing probability. The product $f_e P_e$ is high when expert e is both frequently selected and assigned high probability, penalizing overload. λ_{sw}=0.033 is tuned via hyperparameter optimmization (Appendix A.2).

$\mathcal{L}_{\text{aux}}$ is applied exclusively in the outer loop. During inner-loop adaptation on a new subject's support set, the gate is free to specialize toward whichever subset of experts is most useful for that task. The outer loop then regularizes the meta-initialization toward a balanced routing prior, preserving expert diversity for any new subject at the start of adaptation. This placement performed best during hyperparameter optimization (Appendix A.2).

4 Experiments

4.1 Dataset and Preprocessing

Dataset. We performed a secondary analysis of Yamagami et al. [27], a dataset of 32 participants, 26 with upper-body motor impairments (13 spinal cord injury, 3 muscular dystrophy, 3 peripheral neuropathy, 2 essential tremor, and 5 other disability), recorded using 16 surface EMG electrodes and 12 6-axis IMU sensors spanning the upper body. Participants performed 10 repetitions of 10 standardized gesture prompts, executed in whatever manner was comfortable and accessible to them. While gesture *labels* are shared across participants, gesture *execution* is user-defined, creating high inter-subject variability and motivating the personalization approach developed here. For full details refer to Yamagami et al. [27].

Preprocessing. Raw EMG signals (2000 Hz) were processed following [28]: a 40 Hz fourth-order Butterworth high-pass filter, half-wave rectification, and a 40 Hz low-pass filter, which was downsampled to 20 Hz using a 100 ms sliding window with 50 ms overlap. IMU signals (148 Hz) were downsampled with no additional filtering. This yields synchronized multimodal windows at 20 Hz with C=88 input channels (16 EMG + 12×6 IMU). We apply per-channel mean subtraction followed by cross-channel standard deviation normalization within each modality, preserving relative amplitude relationships across channels while ensuring EMG and IMU signals are comparable in scale [28].

4.2 Evaluation Protocol

All models are evaluated under a unified episodic protocol. Each episode samples N gesture classes uniformly without replacement from a held-out test participant's vocabulary, constructs a support set of NK labeled examples and a disjoint query set of NQ examples (Q=9 per class), runs adaptation on the support set, and measures accuracy on the query set. Results are means over 500 episodes per test participant, averaged across participants under a Leave-Two-Subjects-Out (L2SO) protocol. Our primary condition is N=3, K=1 (1-shot 3-way, chance = 33.3%), as one calibration example per gesture is the most realistic enrollment scenario for new users with motor impairments. We additionally report a $K \in \{1, 3, 5\}$-shot × $N \in \{3, 5, 10\}$-way grid in Section 4.5.

For models trained without meta-learning, test-time adaptation uses gradient-based fine-tuning on the support set: we replace the final classification layer and fine-tune for 10 gradient steps with SGD, matching MAML's adaptation budget. Full fine-tuning consistently outperformed linear probing and is reported throughout (Appendix A.5 shows performance across adaptation step counts). Subject-specific models are trained and evaluated within a single participant using a supervised cross-entropy objective, establishing a lower bound on cross-subject transfer. The external pretrained EMG model [11] requires 2,000 Hz raw EMG input and does not support IMU; it is evaluated on a non-downsampled EMG-only variant and is *not directly comparable* to other conditions.

Statistical Testing. To assess whether performance differences across conditions are statistically significant, we applied a one-way repeated measures ANOVA across all 32 participants, using the pingouin statistical library [23]. Since the same 32 participants contributed accuracy scores under every condition, a repeated measures design is appropriate to account for correlated within-subject observations. Mauchly's test indicated a violation of sphericity ($W = 0.039$, $p = 6.2 \times 10^{-9}$), so degrees of freedom were corrected using the Greenhouse-Geisser method ($\varepsilon = 0.46$). Post hoc paired t-tests were conducted for each ablation against EncoderMoE, with Holm–Bonferroni correction applied across comparisons [10]. Effect sizes are reported as Cohen's d_z [5], the paired-samples effect size computed on per-subject difference scores. The Meta Pretrained EMG condition was excluded from statistical comparisons as it was evaluated on a non-comparable dataset variant (see Section 4.4).

4.3 Hyperparameter Optimization

We optimized hyperparameters using Optuna [1] with a Tree-structured Parzen Estimator sampler, run for 100 trials with a fixed random seed. The objective is mean 1-shot 3-way episodic validation accuracy over 100 episodes on a held-out validation set of N=4 users, drawn from a fixed train/val/test 24/4/4 user split. The test split was not used during hyperparameter optimization. The full search space and chosen values are provided in Appendix A.2.

In addition to hyperparameter tuning over the number of experts, we subsequently conducted an **expert count sweep**, fixing all other hyperparameters and varying only the number of experts E. Results (Appendix A.4) show that performance peaks around E≈24 and is robust within $E \in [8, 24]$ before declining, suggesting limited sensitivity to exact expert count. [1]

4.4 Ablation Models

We isolate the contributions of MAML++ and MoE via parameter-matched ablations. All non-MoE ablations use a single CNN encoder with filter counts expanded to match the total parameter budget of

[1]The code has been anonymized and made public at: https://anonymous.4open.science/r/encoder-moe-ANON/README.md

Table 1: 1-shot, 3-way EMG gesture recognition mean test accuracy (%) evaluated in a L2SO (Leave Two Subjects Out) fashion (mean ± std). † Evaluated on the 2,000 Hz dataset variant; not directly comparable to other conditions (see text). Cohen's d_z [5] measures effect size relative to EncoderMoE across 32 participants (paired); higher values indicate larger performance gaps. d_z is not reported for the Meta Pretrained EMG condition as it is not directly comparable.

Model	**Accuracy (%)**	**Cohen's d_z**
EncoderMoE (ours)	86.7 ± 8.3	—
MoE, No MAML	76.8 ± 8.6	1.17
MAML, No MoE	81.7 ± 9.5	0.56
No MAML, No MoE	78.4 ± 8.3	1.00
Subject-Specific Transfer Learning	72.2 ± 9.9	1.58
Subject-Specific EncoderMoE	64.6 ± 9.4	2.49
Meta Pretrained EMG, Linear Probing	62.1 ± 9.8	†
Meta Pretrained EMG, Full Finetuning	56.0 ± 10.2	†

the expert bank. All ablations share EncoderMoE's hyperparameters to ensure observed differences reflect architecture rather than tuning.

No MAML, No MoE supervised CNN-LSTM, cross-subject. Lower bound on performance.

MoE only, No MAML MoE encoder under supervised training. Isolates routing benefit absent meta-learning.

MAML, No MoE parameter-matched single CNN with MAML++. Critical control: any advantage of EncoderMoE over this baseline is attributable specifically to routing and specialization.

Subject-Specific EncoderMoE EncoderMoE trained within a single subject. Isolates MoE benefit when only one signal profile is present.

Subject-Specific Transfer Learning supervised CNN-LSTM trained within a single subject. Isolates the effect of cross-subject training.

Meta Pretrained EMG evaluated with both linear probing and full fine-tuning. Reference point for large-scale pretraining; not directly comparable (EMG-only, 2kHz input).

4.5 Performance Results

EncoderMoE achieves $86.7 \pm 8.3\%$, outperforming all ablations. The omnibus repeated measures ANOVA was significant ($F(7, 217) = 108.6$, $p_{\text{GG}} = 2.9 \times 10^{-32}$, $\eta^2 = 0.54$), and post hoc paired t-tests confirmed that EncoderMoE significantly outperformed all five directly comparable ablation conditions after Holm–Bonferroni correction (all $p_{\text{corr}} < 1.7 \times 10^{-5}$, Cohen's $d_z \in [0.56, 2.49]$).

Contribution of MAML. Removing MAML while retaining MoE drops accuracy by 9.9 points (76.8% vs. 86.7%), confirming that meta-learned initialization is necessary and not redundant with routing.

Contribution of MoE. The param-matched single-encoder MAML baseline achieves $81.7 \pm 9.5\%$, which is 5.0 percentage points below EncoderMoE. Since parameter counts are matched, this gap is attributable to routing and representational specialization.

Cross-subject vs. subject-specific. Cross-subject EncoderMoE (86.7%) substantially outperforms its subject-specific counterpart (64.6%), confirming that meta-training over diverse users provides a stronger prior than within-subject training alone. Subject-specific transfer learning (72.2%) outperforms subject-specific EncoderMoE (64.6%), which is expected: in the single-subject setting, MoE offers no benefit since there is only one signal morphology to represent; additionally, these hyperparameters were tuned for the cross-subject regime.

Comparison to pretrained model. On the 2,000 Hz EMG-only dataset variant, Kaifosh and Reardon [11] achieves 62.1% with linear probing and 56.0% with full fine-tuning. That linear probing outperforms full fine-tuning is consistent with the pretrained weights already encoding generalizable representations across 4,800 users: fine-tuning overwrites this prior and overfits to the single support example. Both results are well below our approach, suggesting that scale alone is insufficient for user-defined gesture recognition in motor-impaired populations.

Shot $\times$ Way Grid EncoderMoE's advantage over MAML, No MoE is largest at $K{=}1$, precisely the target deployment regime. At $N{=}3$, $K{=}1$, EncoderMoE leads by 3.8 points (88.4% vs. 84.6%); at $N{=}10$, $K{=}1$, the gap widens to 12.1 points (67.7% vs. 55.6%). At higher shot counts, MAML, No MoE closes the gap and matches or slightly exceeds EncoderMoE in several high-K, high-N conditions (Table 4). This is expected: with sufficient labeled examples, sufficient adaptation alone can fit the distribution well, compensating for the absence of routing. Note that these results differ slightly from the results in Table 1 due to differences in evaluation splits (more details in Appendix A.6).

4.6 MoE Routing Analysis

To assess whether EncoderMoE's routing reflects meaningful variation in user physiology, we collect gate weight vectors from 100 query-set episodes per user across all 32 users, using the meta-initialized model prior to inner-loop adaptation. Per-user mean gate weight vectors $\bar{\mathbf{w}}_i \in \mathbb{R}^E$ are averaged over all query samples attributed to user u_i.

Load Balance. With $E{=}22$ experts and top-$k{=}9$, the ideal selection frequency is $9/22 \approx 40.9\%$. The observed range is 17.3% - 65.1% (imbalance ratio $3.8\times$): this imbalance ratio confirms the router has learned a meaningful preference ordering over experts rather than degenerating to uniform selection or single-expert collapse [20, 8]. Full diagnostics are in Appendix A.7.

Ability-Level Routing. We separated users into ability-level groups based on their fine motor control, defined as their ability to touch their thumb and index finger together voluntarily [16]. We define three groups: 1) Non-disabled ($N = 6$), our control group, all of whom can complete this fine motor task, 2) Disabled Fine Motor ($N = 11$), the users with motor impairments who can complete this task, and 3) Disabled No Fine Motor ($N = 15$), the users with motor impairments who cannot complete this task. To test whether routing reflects meaningful user variation, we visualize the *gate mean contrast*: for each group g, the difference between the group's mean gate weight vector (averaged across the group's users) and the global mean (averaged across all users), $\Delta_g = \text{mean}_{i \in g}(\bar{\mathbf{w}}_i) - \bar{\mathbf{w}}$. Gate mean contrast is the appropriate statistic here because our ability-level groups represent a nominal categorical variable; a correlation-based measure would impose an unjustified ordinal structure on the three groups.

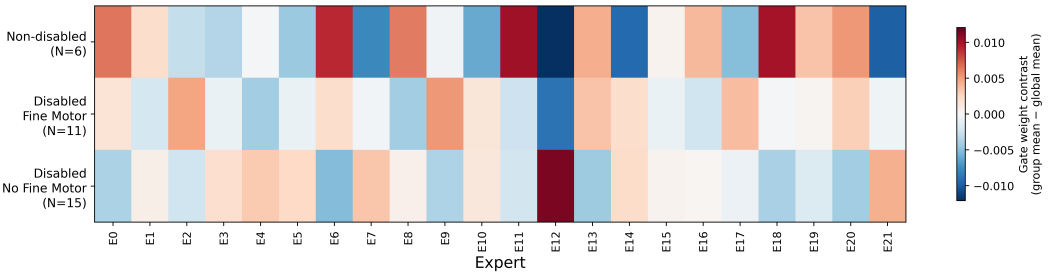


Figure 2: Gate mean contrast heatmap: per-ability-level-group mean gate weight vector minus the global mean, across $E{=}22$ experts. Positive (red) values indicate experts were activated more than average by that group; negative (blue) values indicate below-average activation. The Non-disabled and Disabled Fine Motor groups share similar routing profiles, while the Disabled No Fine Motor group is systematically anti-correlated with those experts, consistent with their more heterogeneous gesture execution strategies. The three ability groups differ in their routing profiles despite the gate receiving no categorical labels at inference time (see Section 4.6).

The heatmap of Figure 2 reveals clear and group-consistent routing differences: experts that activate for the Non-disabled and Disabled Fine Motor groups tend to activate together (i.e., for these rows, the cells are both negative (blue) or both positive (red)), while the Disabled No Fine Motor group is often inversely associated with those experts. In general, users in the Non-disabled and Disabled Fine Motor groups tend to have similar ability levels and thus similar gesture actions, whereas the Disabled No Fine Motor group is much more heterogeneous, frequently employing unique gesture actions [27]. This is immediately clear with expert $E12$, whose gate weights are strongly correlated with the Disabled No Fine Motor users and have a clear negative correlation to the Non-disabled and Disabled Fine Motor groups; this pattern is shared with $E1$ and $E4$. Likewise, some experts learn the reverse pattern, activating the most for the Non-Disabled and Disabled Fine Motor groups, while being inactive for the Disabled No Fine Motor group ($E13$, $E20$, $E6$, $E0$). Critically, these routing differences emerge without any ability label being provided at inference time: the gate is conditioned solely on the learned EMG-IMU context vector, suggesting that the MoE encoder discovers a latent neuromuscular representation that aligns with motor function ability levels. This analysis is exploratory given the group sizes.

5 Conclusion

User-defined gesture recognition for motor-impaired populations sits at the intersection of three challenges that prior work has addressed only in isolation: few-shot cross-subject generalization, semantic heterogeneity in gesture execution, and the highly idiosyncratic signal profiles of diverse motor disabilities. EncoderMoE addresses all three jointly, and our ablations isolate why the combination is necessary: meta-learning alone cannot compensate for the representational mismatch between a global encoder and a heterogeneous population, while routing alone lacks the initialization needed to adapt from a single support example.

Beyond accuracy, the routing analysis reveals that the model discovers ability-level structure from raw biosignals alone, without access to disability labels. This emergent organization suggests that the heterogeneity we sought to accommodate is genuinely latent in the signal, and that the MoE routing mechanism is recovering it rather than ignoring it.

Limitations. Several limitations should be acknowledged. First, our evaluation is conducted on a single dataset of $N{=}32$ participants; results may not generalize to different sensor configurations, tasks, or populations. Second, our dataset is uniquely challenging (user-defined gesture semantics, high disability heterogeneity): we make no claims about generalization to standard fixed-gesture EMG benchmarks where the inductive biases of MAML and MoE may be less necessary. Third, performance degrades substantially in the 10-way regime (67.7% at 1-shot 10-way) with limited recovery at higher shot counts, suggesting the current architecture does not scale gracefully to large gesture vocabularies.

Broader Impacts. This work targets a direct accessibility application: reducing calibration burden for users with motor impairments interacting with assistive devices or XR interfaces via personalized gesture vocabularies. Any system capturing biosignal data carries privacy risk, such as re-identification from raw signals or model weights [14]: deployment should be accompanied by informed consent protocols and data minimization practices.

Future Extensions. Three directions are most immediate. First, scaling to larger gesture vocabularies ($N \geq 10$) remains an open challenge; architectural changes such as hierarchical routing or vocabulary-aware episode construction during meta-training may be necessary. Second, the interaction between MoE gating and MAML's inner-loop adaptation is not yet theoretically characterized. The critical feature for MoE is the gating mechanism, whereas for MAML the inner-loop adaptation likely updates the classification head: understanding whether meta-learning primarily specializes routing, representations, or both could inform more principled architecture design.

References

[1] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama. Optuna: A next-generation hyperparameter optimization framework. In *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*, pages 2623–2631, 2019.

[2] A. Antoniou, H. Edwards, and A. Storkey. How to train your maml. *arXiv preprint arXiv:1810.09502*, 2018.

[3] G. Balbinot, G. Li, M. J. Wiest, M. Pakosh, J. C. Furlan, S. Kalsi-Ryan, and J. Zariffa. Properties of the surface electromyogram following traumatic spinal cord injury: a scoping review. *Journal of neuroengineering and rehabilitation*, 18(1):105, 2021.

[4] P. Chen, S. Wang, S. Hu, X. Wu, Y. Shu, Z. Rao, M. Wang, Y. Li, B. Yang, and C. Guo. Empowering time series analysis with large-scale multimodal pretraining. *arXiv preprint arXiv:2602.05646*, 2026.

[5] J. Cohen. *Statistical power analysis for the behavioral sciences*. routledge, 2013.

[6] J. G. Colli Alfaro and A. L. Trejos. User-independent hand gesture recognition classification models using sensor fusion. *Sensors*, 22(4):1321, 2022.

[7] U. Côté-Allard, C. L. Fall, A. Drouin, A. Campeau-Lecours, C. Gosselin, K. Glette, F. Laviolette, and B. Gosselin. Deep learning for electromyographic hand gesture signal classification using transfer learning. *IEEE transactions on neural systems and rehabilitation engineering*, 27(4): 760–771, 2019.

[8] W. Fedus, B. Zoph, and N. Shazeer. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research*, 23(120):1–39, 2022.

[9] C. Finn, P. Abbeel, and S. Levine. Model-agnostic meta-learning for fast adaptation of deep networks. In *International conference on machine learning*, pages 1126–1135. PMLR, 2017.

[10] S. Holm. A simple sequentially rejective multiple test procedure. *Scandinavian journal of statistics*, pages 65–70, 1979.

[11] P. Kaifosh and T. R. Reardon. A generic non-invasive neuromotor interface for human-computer interaction. *Nature*, 645(8081):702–711, 2025.

[12] D. P. Kingma and J. Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.

[13] P. L. La Rotta, J. Xu, A. Chen, L. Winterbottom, W. Chen, D. Nilsen, J. Stein, and M. Ciocarlie. Meta-learning for fast adaptation in intent inferral on a robotic hand orthosis for stroke. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 4693–4700. IEEE, 2024.

[14] K. Malcolm, C. A. Uribe, and M. Yamagami. Federated learning in offline and online emg decoding: A privacy and performance perspective. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 2026.

[15] A. Nichol, J. Achiam, and J. Schulman. On first-order meta-learning algorithms. *arXiv preprint arXiv:1803.02999*, 2018.

[16] A. A. Portnova-Fahreeva, M. Yamagami, A. Robert-Gonzalez, J. Mankoff, H. Feldner, and K. M. Steele. Accuracy of video-based hand tracking for people with upper-body disabilities. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 32:1863–1872, 2024.

[17] K. Proroković, M. Wand, and J. Schmidhuber. Meta-learning for recalibration of emg-based upper limb prostheses. In *4th Lifelong Machine Learning Workshop at ICML 2020*, 2020.

[18] F. Rizzoglio, C. Pierella, D. De Santis, F. Mussa-Ivaldi, and M. Casadio. A hybrid body-machine interface integrating signals from muscles and motions. *Journal of Neural Engineering*, 17(4): 046004, 2020.

[19] S. Shahi, V. Mollyn, C. Tymoszek Park, R. Kang, A. Liberman, O. Levy, J. Gong, A. Bedri, and G. Laput. Vision-based hand gesture customization from a single demonstration. In *Proceedings of the 37th annual ACM symposium on user interface software and technology*, pages 1–14, 2024.

[20] N. Shazeer, A. Mirhoseini, K. Maziarz, A. Davis, Q. Le, G. Hinton, and J. Dean. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*, 2017.

[21] J. Snell, K. Swersky, and R. Zemel. Prototypical networks for few-shot learning. *Advances in neural information processing systems*, 30, 2017.

[22] C. Tong, J. Ge, and N. D. Lane. Zero-shot learning for imu-based activity recognition using video embeddings. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 5(4):1–23, 2021.

[23] R. Vallat. Pingouin: statistics in python. *J. Open Source Softw.*, 3(31):1026, 2018.

[24] O. Vinyals, C. Blundell, T. Lillicrap, D. Wierstra, et al. Matching networks for one shot learning. *Advances in neural information processing systems*, 29, 2016.

[25] Y. Wu and K. He. Group normalization. In *Proceedings of the European conference on computer vision (ECCV)*, pages 3–19, 2018.

[26] X. Xu, J. Gong, C. Brum, L. Liang, B. Suh, S. K. Gupta, Y. Agarwal, L. Lindsey, R. Kang, B. Shahsavari, et al. Enabling hand gesture customization on wrist-worn devices. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, pages 1–19, 2022.

[27] M. Yamagami, A. A. Portnova-Fahreeva, J. Kong, J. O. Wobbrock, and J. Mankoff. How do people with limited movement personalize upper-body gestures? considerations for the design of personalized and accessible gesture interfaces. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility*, pages 1–15, 2023.

[28] M. Yamagami, C. L. Mitchell, A. A. Portnova-Fahreeva, J. Kong, J. Mankoff, and J. O. Wobbrock. Customized mid-air gestures for accessibility: A $b recognizer for multi-dimensional biosignal gestures. *arXiv preprint arXiv:2409.08402*, 2024.

[29] J. Yang, K. Shibata, D. Weber, and Z. Erickson. High-density electromyography for effective gesture-based control of physically assistive mobile manipulators. *npj Robotics*, 3(1):2, 2025.

A Appendix

A.1 Model-Agnostic Meta-Learning (MAML)

MAML [9] seeks meta-parameters θ that adapt rapidly to any new task $\mathcal{T}_i$ via a small number of gradient steps on the support set. The *inner loop* produces task-specific parameters ϕ_i by taking M gradient steps from θ:

$$\phi_i^{(m)} = \phi_i^{(m-1)} - \boldsymbol{\alpha}^{(m)} \odot \nabla_{\phi_i^{(m-1)}} \mathcal{L}^{\mathrm{spt}}\left(\phi_i^{(m-1)}\right), \quad m = 1, \ldots, M, \tag{4}$$

where $\phi_i^{(0)} = \theta$ and $\boldsymbol{\alpha}^{(m)} \in \mathbb{R}^{|\theta|}$ is a vector of learned per-parameter step sizes at inner step m (MAML++ [2] learned step sizes; vanilla MAML uses a shared scalar α). The *outer loop* updates θ by minimizing the query-set loss across tasks, accumulated over all inner steps with scheduled weights ω_m (MAML++ multi-step loss):

$$\theta \leftarrow \theta - \beta \nabla_\theta \sum_{\mathcal{T}_i \sim p(\mathcal{T})} \sum_{m=1}^{M} \omega_m \, \mathcal{L}^{\mathrm{qry}}\left(\phi_i^{(m)}\right), \tag{5}$$

where β is the outer learning rate.

A.2 Hyperparameter Optimization Details

Table 2 reports the full search space and the value selected by hyperparameter optimization for each tuned parameter. Table 3 reports the parameters that were fixed throughout.

Table 2: Hyperparameter search space and selected values for EncoderMoE. All values are from the best trial, which achieved 90.1% mean 1-shot 3-way validation accuracy on the fixed validation split. Validation accuracy exceeds reported test accuracy (Table 1) as hyperparameter optimization was optimized on the validation split, whereas the test accuracy is reported as L2SO. "log-unif" denotes log-uniform sampling; "cat" denotes categorical. Parameters are grouped by the component they govern.

Group	Parameter	Type	Search range / options	Chosen value
MAML optimization	Outer LR β	log-unif	$[5\times10^{-5},\ 5\times10^{-4}]$	1.95×10^{-4}
	Weight decay	log-unif	$[1\times10^{-5},\ 1\times10^{-3}]$	8.87×10^{-4}
	Inner steps M	cat	$\{5, 10, 15, 20, 25\}$	10
	Inner LR α_{init}	log-unif	$[1\times10^{-4},\ 1\times10^{-2}]$	9.73×10^{-4}
	Eval LR α_{eval}	log-unif	$[1\times10^{-4},\ 5\times10^{-2}]$	5.07×10^{-3}
Regularization	Label smoothing ε	cat	$\{0.0, 0.05, 0.1, 0.15\}$	0.05
	MoE dropout	log-unif	$[1\times10^{-2},\ 2\times10^{-1}]$	0.037
MoE structure	Num. experts E	cat	$\{4, 8, 12, 16, 20, 24, 28, 32\}$	22
	Top-k utilization ratio	cat	$\{0.1, 0.2, 0.3, 0.4, 0.5\}$	0.41 (k=9)
	Gate temperature τ	log-unif	$[0.3,\ 3.0]$	1.53
	Context hidden dim	cat	$\{32, 64, 128\}$	32
	Context output dim	cat	$\{16, 32, 64\}$	64
MoE loss	Aux. loss coeff. λ_{sw}	log-unif	$[1\times10^{-3},\ 1\times10^{-1}]$	0.033
	Aux. loss placement	cat	{inner, outer, both}	outer
Training schedule	Episodes per epoch	cat	$\{200, 300, 400, 500\}$	500

Table 3: Fixed hyperparameters for EncoderMoE. These were either determined during the first HPO stage and fixed in the second, or fixed by architectural design throughout.

Group	Parameter	Value
Architecture	CNN base filters F_{base}	64 (3 layers: 64, 128, 256 filters)
	LSTM hidden size H	64 (bidirectional; output dim 128)
	GroupNorm groups	8
	Global average pooling	Enabled
MAML configuration	LSLR (learned per-param step sizes)	Enabled (train); disabled (eval)
	Multi-step loss (MSL) schedule	Enabled
	Gradient order	First-order (fo-MAML)
	Meta-batch size B_{meta}	24 tasks
	Outer optimizer	Adam
MoE configuration	Expert placement	Encoder (parallel to CNN)
	Shared (always-on) expert	Disabled
	Routing signal	Context projector (`context_proj`)
Training schedule	Total epochs	23
	Val. episodes (final ablation runs)	200
	Test episodes	500
	Gradient clip max norm	10.0
Task definition	N-way / K-shot / Q-query	3-way / 1-shot / 9-query

A.3 Computational Resources

All experiments were conducted on the NOTS cluster at Rice University's Center for Research Computing (CRC), using NVIDIA Tesla V100 and A40 GPUs. Each job used a single GPU, 10 CPU cores, and between 16–32 GB of RAM depending on the ablation. Our total compute across the full project (including hyperparameter optimization, exploratory runs, and all final reported experiments) was approximately 5,800 GPU-hours.

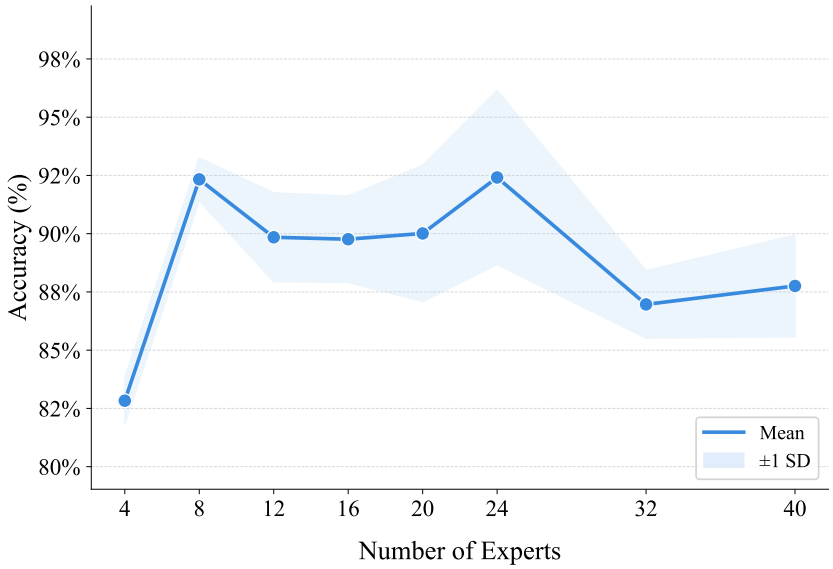


Figure 3: 1-shot 3-way accuracy as a function of expert count E, with all other hyperparameters fixed at EncoderMoE's optimal configuration. Evaluated on the fixed test split from hyperparameter optimization.

A.4 Number of Experts Sweep

Figure 3 plots 1-shot 3-way accuracy as a function of the number of experts E, with all other hyperparameters fixed at our proposed MAML+MoE's optimal configuration. The peak at E=24 suggests that the dataset's inter-subject variability is best captured by a pool of 24 specialized sub-encoders; beyond this point, the routing problem becomes harder and individual experts are under-utilized. We note that the within-trial inter-subject variability is higher than the difference between expert-varying trials: thus, while $E = 24$ does appear to be the peak, our model appears to be robust to a wide number of experts, but we note a clear drop off starting at 4 or less experts, as well as a drop off when using more than 32 experts.

A.5 Finetuning and Adaptation Sweep

Figure 4 shows accuracy as a function of adaptation step count for EncoderMoE and the No MAML, No MoE baseline, representing MAML-based and standard fine-tuning adaptation procedures, respectively. Both models peak at approximately 10 steps before plateauing, providing convergent evidence that 10 adaptation steps are sufficient regardless of the underlying adaptation mechanism. We report No MAML, No MoE rather than MAML, No MoE here because the two MAML-based models share identical inner-loop adaptation, and the more informative comparison is between the two adaptation regimes (meta-learned vs. fine-tuning); the agreement between them set 10 steps as our selected budget.

A.6 Shot x Way Grid

Table 4 reports accuracy across a $K \in \{1, 3, 5\} \times N \in \{3, 5, 10\}$ grid for EncoderMoE and the best directly comparable baseline (MAML, No MoE). All models use hyperparameters tuned for the 1-shot 3-way condition and are trained from scratch at each shot–way configuration; no additional hyperparameter optimization was performed.

EncoderMoE's advantage is largest in the low-data regime. At K=1, EncoderMoE leads by 3.8 points at N=3 and 12.1 points at N=10. The intuition is that with only a single support example, MAML's inner loop has minimal gradient signal to adapt from, so the routing prior (the model's learned ability to identify which experts are relevant for a given user from the EMG-IMU context

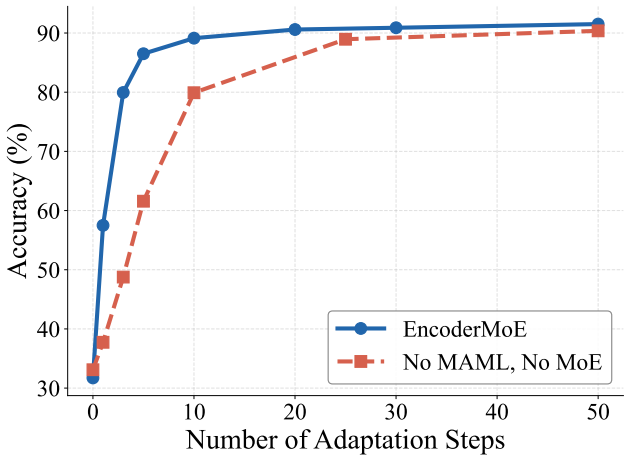


Figure 4: Accuracy as a function of the number of inner-loop adaptation steps at meta-test time for EncoderMoE and the No MAML, No MoE baseline.

alone) carries most of the adaptation burden. MoE is well-suited to this regime precisely because routing operates directly on the query input and does not depend on the support set size.

As K increases, MAML, No MoE closes the gap and matches or exceeds EncoderMoE in several high-K, high-N conditions (5-shot 5-way: 89.4% vs. 89.0%; 3-shot 10-way: 68.4% vs. 64.4%). With more labeled examples available, gradient-based adaptation can compensate for the absence of routing, reducing the marginal benefit of the MoE prior. This pattern is consistent with our primary claim: EncoderMoE's routing is most valuable precisely in the 1-shot regime that defines our target deployment scenario.

Table 4: Accuracy (%) across shot $\times$ way conditions for EncoderMoE and the best baseline (MAML, No MoE). All models use the 1-shot 3-way hyperparameter configuration without additional hyperparameter optimization, the only change being that models are trained from scratch with an updated K-shot N-way. Models are evaluated on the withheld test set of the fixed train/val/test split from hyperparameter optimization. **Bold** indicates the better result for each shot–way condition.

		Ways (N)		
Model	**Shots** (K)	3	5	10
EncoderMoE (ours)	1	$\mathbf{88.4 \pm 2.9}$	$\mathbf{78.2 \pm 3.8}$	$\mathbf{67.7 \pm 9.6}$
	3	$\mathbf{95.7 \pm 1.7}$	$\mathbf{89.0 \pm 5.0}$	64.4 ± 10.3
	5	$\mathbf{96.9 \pm 2.0}$	89.0 ± 3.3	68.5 ± 10.3
MAML, No MoE	1	84.6 ± 1.2	75.9 ± 3.4	55.6 ± 4.1
	3	94.7 ± 2.0	87.9 ± 4.0	$\mathbf{68.4 \pm 4.7}$
	5	96.4 ± 2.2	$\mathbf{89.4 \pm 4.0}$	$\mathbf{69.9 \pm 2.7}$

A.7 Load-Balance Diagnostics

We report two complementary metrics of expert utilization computed over all 21,600 query-set samples pooled across 32 participants, using the meta-initialized EncoderMoE prior to inner-loop adaptation.

Mean soft gate weight per expert. For each expert e we average the post-softmax gate weight $w_e^{(b)}$ across all samples. The observed soft weights range from 0.107 to 0.653 (imbalance ratio $6.1\times$).

Expert selection frequency (top-k binary). We binarize each routing decision and compute the fraction of samples for which each expert is selected. Selection frequency ranges from 17.3% to 65.1% (imbalance ratio $3.8\times$); no expert has zero frequency.

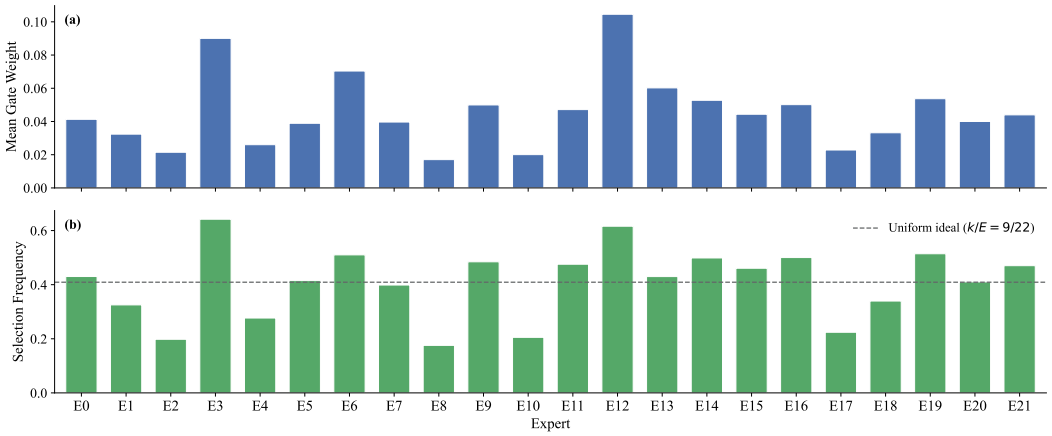


Figure 5: **Expert routing statistics across samples.** (a) Mean soft gate weight per expert, i.e., the average of the continuous gating score assigned to each expert by the router across all input samples. (b) Hard selection frequency per expert under top-k routing, i.e., the fraction of samples for which each expert falls within the top-k selected set. The dashed line in (b) denotes the uniform ideal (k/E), the expected frequency if the router distributed selections perfectly evenly.

Taken together, these two metrics bracket the routing behavior. Pathological degenerate routing would manifest as either uniform selection (imbalance ratio $\approx 1\times$, indicating the router has learned nothing) or single-expert collapse (one expert selected for all samples). Neither is observed here. The moderate hard-selection imbalance ($3.8\times$) confirms the router has learned a meaningful preference ordering, while the absence of zero-frequency experts confirms no expert is dead or ignored. The larger soft-weight imbalance ($6.1\times$) relative to hard-selection imbalance ($3.8\times$) reflects the continuous nature of the gating scores: experts that are frequently selected also tend to receive disproportionately high weight within the top-k set, suggesting the router assigns not just binary membership but also a confidence-weighted contribution to each selected expert.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The claims in the abstract and introduction directly match the paper's main contribution and comparison against our ablations.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: We include a Limitations subsection in Conclusion, primarily detailing 1) experimental constraints (low number of subjects) and 2) dataset constraints (we have a unique dataset for user-defined gestures and thus only train/evaluate on this dataset).

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
   - While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: This work does not include theoretical results.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: We describe our hyperparameter optimization route, our train/test split, and our model architecture.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Code has been anonymized and made publicly available at `https://anonymous.4open.science/r/encoder-moe-ANON/README.md`. The dataset is not publicly available as it was collected by a third party; we cite the original authors.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: The Methods and Appendix detail the architecture and hyperparameter tuning procedure such that the model can be replicated.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: For our main results and ablations, we conduct Leave Two Subjects Out. Comparing against ablations, we run a one-way repeated measures ANOVA and show that our EncoderMoE outperforms all ablations in a statistically significant manner.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.

- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Full information can be found in A.3.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The research conducted conforms with the NeurIPS Code of Ethics.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [Yes]

    Justification: We include broader impacts in our Conclusion. We cover the positive societal impact of making gesture classification and biosignal-based human-computer interaction more accessible, while also acknowledging privacy risks.

    Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

    Justification: Our dataset is small, and our model is only suited for a very specific task (EMG-IMU user-defined gesture classification). We do not anticipate any risks with our model or code, given the small scale of our dataset.

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.
    - Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
    - Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
    - We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: We credit and cite the group that collected the dataset that we used in the paper; at the time of submission, that dataset is not publicly available. Our code is released under the MIT License at `https://anonymous.4open.science/r/encoder-moe-ANON/README.md`.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.

- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: Code has been anonymized and made publicly available at `https://anonymous.4open.science/r/encoder-moe-ANON/README.md`, with a README describing usage. Released under the MIT License.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: While we do use a human subjects dataset that was collected under IRB approval, we were not the ones who collected the dataset, as this work is a secondary analysis of said dataset. All study instructions and protocols can be found in the papers that collected the dataset: [27, 28]

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: While we do use a human subjects dataset that was collected under IRB approval, we were not the ones who collected the dataset, as this work is a secondary analysis of said dataset. All IRB-related information is contained in the papers that collected the dataset: [27, 28]

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [N/A]

Justification: LLMs were not used in the development of the core method.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.